Hyeongyu Kim

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GitHub | LinkedIn | Google Scholar

Test-time adaptation / Reliable machine learning / Efficient deployment

Research in test-time adaptation and calibration under distribution shift, with current compiler work supporting efficient model deployment.

Experience

Hyundai Motor Company — South Korea
Compiler Engineer, Compiler Team — *Aug. 2026 – Present*

- Develop TVM-based compilation for custom-NPU deployment (Relax/TIR lowering, tiling, scheduling, memory mapping), with HW/SW co-design to balance latency, bandwidth, throughput, and power.

Yonsei University, Medical Artificial Intelligence Lab — Seoul, South Korea
Doctoral Researcher; Advisor: Prof. Dosik Hwang — *Mar. 2020 – Aug. 2026*

- Developed activation-aware test-time adaptation by parameterizing ReLU/GELU behavior; evaluated robustness on CIFAR-10-C, CIFAR-100-C, and ImageNet-C (AcTTA, CVPR 2026, co-first author).
- Designed modular buffer layers for online adaptation while keeping backbone weights fixed (NeurIPS 2025, co-first author); co-developed complementary additive/subtractive adaptation for adverse-weather detection (CVPR 2026 Highlight).
- Developed mutual-information-guided calibration and adaptation using cached features and classifier perturbations (MIRA, NeurIPS 2026 accepted, co-first author).
- Built a volumetric cross-modality adaptation pipeline without target labels; the team ranked 1st among 16 teams in CrossMoDA 2021 (CVPR 2023; MIA benchmark).

Education

Yonsei University — Seoul, South Korea
Ph.D., Electrical and Electronic Engineering; Fellow Scholarship — Mar. 2020 – Aug. 2026
B.S., Electrical and Electronic Engineering — Mar. 2014 – Feb. 2020

Technical Skills

Research: Test-time adaptation, calibration, domain adaptation/generalization, 3D segmentation
Languages / ML: Python, C/C++, MATLAB; PyTorch, TensorFlow, CUDA, OpenCV
Deployment / systems: TVM (Relax, TIR), tiling, scheduling, memory mapping, Git, Docker, Linux

Selected Publications

* Equal contribution. Full publication list: Google Scholar.

- **MIRA: Mutual Information guided calibration for Reliable Test-Time Adaptation**. H. Kim*, M. Jang*, Y. Song, G. Han, D. Hwang. *NeurIPS 2026, accepted.*
- **AcTTA: Rethinking Test-Time Adaptation via Dynamic Activation**. H. Kim*, G. Han*, D. Hwang. *CVPR 2026.*
- **CD-Buffer: Complementary Dual-Buffer Framework for Test-Time Adaptation in Adverse Weather Object Detection**. Y. Song, H. Kim, D. Hwang. *CVPR 2026, Highlight.*
- **Buffer layers for Test-Time Adaptation**. H. Kim*, G. Han*, D. Hwang. *NeurIPS 2025.*
- **Automatic segmentation and modeling of the aortic vessel tree: Overview of the SEG.A 2023 aorta segmentation challenge**. Y. Jin et al. (including H. Kim). *Medical Image Analysis, online first 2026; doi: 10.1016/j.media.2026.104324.*

Selected Projects & Recognition

- **Outstanding Reviewer, CVPR 2026.**
- **Samsung Medison / Severance Hospital (2025):** Developed fetal ultrasound enhancement using diffusion and flow matching for denoising, super-resolution, and artifact suppression.
- **Samsung Electronics (2023):** Developed an MRI-based Alzheimer's disease diagnosis algorithm; patent application recorded as DP-2023-0002.
- **CrossMoDA 2021:** 1st place. **2021 medical AI competitions:** Two Excellence Awards. **Hyundai Motor Group:** Research Scholarship (2024). Five patent applications (2023–2025).